

The Status of Research in Localisation

In this 10th and concluding article in the series on localisation, the author take a closer look at the challenges for Indian language localisation and the current status of research in the field.



Ten years back, there were very few people who knew how to use Indian languages on the desktop. Now, we not only have millions of people who read content on the Internet in their native languages on desktops and smartphones, but also thousands of people who edit and contribute content. This has been possible due to the availability of computing devices with Indian language support, their falling prices, as well as the ability to access the Internet through various communication mediums.

We have seen a few early signs of the outcome of the research of the last decade, in the form of machine translation support for the Web. While five Indian languages are supported by Google's machine translation tools, the quality is still not up to the mark due to the complex nature of languages. Speech and touch interfaces have made their appearance, particularly on smartphones. The speech interface is now supported in limited domains, such as searching through the contacts list, or searching the Internet. However, Indian-accented English support needs to be improved.

As per the framework of the Centre for Next Generation Localisation, a specialised centre of excellence in Ireland, the challenges for localisation are volume, access (interaction mode) and personalisation. These three dimensions represent three axes, with most of the localisation work focused on high-volume content in corporate environments, with support for

the traditional keyboard-and-screen mode of access, and limited support for personalisation in terms of language variations. The research challenge is to leverage various core technologies and frameworks to be able to instantly translate content and localise applications, duly considering the profile of the user. The next few paragraphs explore the status of localisation and the challenges faced in dealing with Indian languages.

a) Volume: The quantity of information on the Web is exploding due to its popularity as a medium of communication and interaction, and also the popularity of Web 2.0 platforms such as Twitter, Facebook, Google+, etc. The industry has tried to address this by defining and improving the process for localisation in corporate environments, as well as leveraging the crowd sourcing opportunity in social media environments.

The core component of localisation is the translation technology. For a long time, the route explored was rule-based translation research consisting of parsing of the source text, and using dictionaries and grammar rules to produce the translation. Subsequently, Statistical Machine Translation(SMT), based on training of the algorithms, with paired human-translated texts of source language text and destination language texts has become popular. Websites, translated at the click of a button for the dominant languages, have become feasible—though the quality of the translation could be inadequate for professional requirements.